

**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**

**B.Tech (Computer Science and Engineering(Artificial Intelligence and Machine Learning)) SEMESTER - 6 Summer 2025 (Regular)**

**Course :Bachelor of Technology (Computer Science and Engineering(Artificial Intelligence and Machine Learning)) Branch : {FACULTYNAME}**

**Semester : SEMESTER - 6**

**Subject Code & Name: BTAIC601 - DEEP LEARNING**

**Time : 3 Hours]**

**[Total Marks : 60**

**Instructions to the Students:**

- 1.Each question carries 12 marks.
- 2.Question No. 1 will be compulsory and include objective-type questions.
- 3.Candidates are required to attempt any four questions from Question No. 2 to Question No.6
- 4.Use of non-programmable scientific calculators is allowed.
- 5.Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

12

How many layers Deep learning algorithms are constructed?

1. 2
2. 3
3. 4
4. 5

2 The first layer is called the

1. inner layer
2. outer layer
3. hidden layer
4. None of the above

3 Which is NOT a component of a biological neuron?

1. Dendrites
2. Axon
3. Synapse
4. Nucleus

4 The McCulloch-Pitts neuron is considered a \_\_\_\_ model of a biological neuron.

1. Simplified
2. Complex
3. Realistic
4. Stochastic

5 What type of neural network is used for image classification tasks?

1. RNN
2. CNN
3. MLP
4. RBFN

6 Which is a common activation function used in the output layer for binary classification?

1. Sigmoid
2. ReLU
3. Tanh
4. Softmax

7 Deep learning is

1. A type of machine learning algorithm
2. A branch of artificial intelligence
3. A technique for training neural networks
4. All of the above

8 The purpose of pooling layers in a convolutional neural network (CNN) is

1. To reduce the size of the input data
2. To perform spatial downsampling
3. To extract the most important features
4. All of the above

9 Which type of neural network architecture is often used for object detection tasks?

1. LSTM
2. GAN
3. CNN
4. RNN

10 Sentiment analysis using Deep Learning is a many-to one prediction task

1. True
2. False
3. can be true & false
4. cannot say

11 Which neural network has only one hidden layer between the input and output?

1. Shallow NN
2. Deep NN
3. Feed-forward NN
4. Recurrent NN

12 RNNs stands for?

1. Recurrent NNs
2. Report NNs
3. Recording NNs
4. Recurrent NNs

Q2. Solve the following.

A) What is Gradient Descent and its variants?

6

B) Differentiate between forward propagation and backpropagation networks.

6

Q3. Solve the following.

- A) Explain Building Blocks of Convolutional Neural Network. 6
- B) List different CNN Architectures and explain any two in detail. 6

Q4. Solve Any Two of the following.

- A) Explain Variational Autoencoder. 6
- B) Discuss transfer learning with advantages, disadvantages and uses. 6
- C) Explain working of Recurrent Neural Network (RNN). 6

Q5. Solve Any Two of the following.

- A) Explain Generative Adversarial Network (GAN) architecture. 6
- B) Explain Early Stopping and Dropout in effective training. 6
- C) Give brief introduction to Sequential Temporal Data. 6

Q6. Solve Any Two of the following.

- A) Explain applications of Deep Learning algorithms. 6
- B) Explain how LSTM network works step-by-step? 6
- C) Explain Building Blocks of Deep NN. 6